

Evaluation of Eulerian and Lagrangian Models for Hemolysis Estimation

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Hemolysis caused by flow-induced mechanical damage to red blood cells is still a problem in medical devices such as ventricular assist devices (VADs), artificial lungs, and mechanical heart valves. A number of different models have been proposed by different research groups for calculating the hemolysis, and of these, the power law-based models ($HI(\%) = Ct^\alpha \tau^\beta$) have proved the most popular because of their ease of use and applicability to a wide range of devices. However, within this power law category of models there are a number of different implementations. The aim of this work was to evaluate different power law-based models by calculating hemolysis in a specifically designed shearing device and a clinical VAD, and comparing the estimated results with experimental measurements of the hemolysis in these two devices. Both the Eulerian scalar transport and all the Lagrangian models had fairly large percentage of errors compared with the experiments (minimum Eulerian 91% and minimum Lagrangian 57%) showing they could not accurately predict the magnitude of the hemolysis. However, the Eulerian approach had large correlation coefficients (>0.99) showing that this method can predict relative hemolysis, which would be useful in comparative analysis, for example, for ranking different devices or for design optimization studies. *ASAIO Journal* 2012; 58:363–372.

Damage to the red blood cell (RBC) with resulting release of hemoglobin into the blood plasma is known as hemolysis. This blood cell trauma manifests as morphologic changes, shortened life span, biochemical functional alterations, and complete rupture. Mechanical shear stress is known to cause hemolysis either by exerting sufficient stress to completely rupture the RBC membrane or through formation of pores in the RBC membrane when the stress/strain exceeds a critical value. In addition to the magnitude of the shear stress, hemolysis is also increased by increasing duration of

the RBC's exposure to shear stress.^{1–3} Shear stress below a critical level results in either elastic reversible elongation of the cells, if applied for a short time, or nonreversible shape change with longer application. The former conditions do not create hemolysis. However, the latter one may yield hemoglobin leakage into the blood plasma.⁴

Two major types of models for hemolysis have been proposed: strain-based models^{5–7} and power law models. Strain-based models aim to model the deformation of the RBCs and corresponding strain of the cell membrane and calculate the subsequent hemoglobin released, a bottom up approach. In contrast, power law models are derived from the empirical observation that under constant uniform shear stress the increase in plasma free hemoglobin, ΔHb , is a power law function of shear stress and exposure time, a top down approach. Power law models have been the most popular, probably because of their ease of implementation and apparent broad applicability. The first power law model was developed by Giersiepen *et al.*,⁸ for use in calculating the hemolysis index (HI), due to mechanical heart valves, and was based on *in vitro* hemolysis data obtained from a Couette shearing device in which human RBC suspension was exposed to shear stresses, τ , (<255 Pa), for exposure times, t (<700 msec):

$$HI(\%) = \frac{\Delta Hb}{Hb} \times 100 = Ct^\alpha \tau^\beta \quad (1)$$

The constants C , α , and β were determined by regression of the experimental data. Heuser and Opitz⁹ used the same shearing device with porcine blood, shorter exposure times ($t < 700$ msec), and higher shear stresses ($\tau < 700$ Pa).

With the aim of developing a more widely applicable power law equation Zhang *et al.*¹⁰ recently conducted uniform shear stress experiments using novel shearing devices. Ovine blood was exposed to uniform shear stresses (50–320 Pa) for a range of exposure times (<1.5 sec), and the results were fit to Equation 1. The resulting empirical constants are given in **Table 1** (Zhang *et al.* constants [TZ]), along with those of Giersiepen (GW) *et al.* and Heuser *et al.* (HO).

Over the last decade there have been persistent attempts to use computational fluid dynamics (CFD) for predicting hemolysis.¹¹ In general, there are two types of computational methods available: Eulerian and Lagrangian approaches. In the Eulerian approach, the damage index is integrated over the entire computational flow domain whereas in the Lagrangian formulation the integration is along the flow path lines.

The Eulerian approach has been used to calculate the blood damage potential of ventricular assist devices (VADs) by several research groups.^{12–14} $\Delta Hb'$ can be defined as a scalar

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Table 1. Power Law Equation Constants and Their Covering Ranges

Model	Range		C	α	β
	Shear Stress (Pa)	Exposure Time (msec)			
GW	<255	<700	3.620×10^{-5}	0.7850	2.4160
HO	<700	<700	1.800×10^{-6}	0.7650	1.9910
TZ	50–320	<1500	1.228×10^{-5}	0.6606	1.9918

GW, Giersiepen *et al.* constants⁸; HO, Heuser *et al.* constants⁹; TZ, Zhang *et al.* constants¹⁰; C, α and β , constants of the power law equation.

variable that is equal to $\Delta Hb^{1/\alpha 15}$ (similar to the method used by Fill *et al.*).¹⁶ The scalar transport equation is then expressed as:

$$\frac{d(\Delta Hb')}{dt} + v\rho \cdot \nabla(\Delta Hb') = S \quad (2)$$

where S is the source term and defined as $S = \rho(Hb \cdot C \cdot \tau^\beta)^{1/\alpha}$. It is possible to use other source terms for plasma free hemoglobin¹⁷ although these have never been applied in practice.

The Lagrangian approach encompasses the ability to calculate the shear stress history on the RBC, which enables the modeling of prior damage. In the Lagrangian approach, path lines are traced from the inlet of a device to the outlet. Although the number of path lines cannot possibly reproduce the number of actual paths taken, these path lines are assumed to be typical paths that are representative of all the paths. Several damage accumulation models have been proposed. These all use the assumption that, although the shear stress varies throughout a medical device, over a short enough time the shear stress can be considered uniform and therefore hemolysis for this time interval can be calculated based on Equation 1. The hemolysis for each time interval can then be integrated along each path line and the average of the hemolysis at the outlet for all the path lines is then found. Several groups^{18–20} considered the hemolysis for the short time interval to be given by Equation 1 and therefore the total hemolysis along a path line from the inlet of a device to the outlet was found by integrating equation 1:

$$HI1 = \int_{\text{inlet}}^{\text{outlet}} C \cdot dt_{\text{exp}}^\alpha \cdot \tau^\beta = \sum_{\text{inlet}}^{\text{outlet}} C \cdot \Delta t_{\text{exp}}^\alpha \cdot \tau^\beta \quad (3)$$

As noted by Grigioni *et al.*²¹ this equation fails to estimate the uniform shear stress hemolysis as it does not reproduce the total hemolysis given by Equation 1.

Several other groups^{22–24} calculated the accumulated damage by integrating the temporal derivative of Equation 1:

$$HI2 = \sum_{\text{inlet}}^{\text{outlet}} \alpha \cdot C \cdot t_{\text{exp}}^{\alpha-1} \cdot \tau^\beta \cdot \Delta t_{\text{exp}} \quad (4)$$

$HI2$ integrates to give Equation 1 for a uniform shear stress field but does not incorporate the damage history.

Based on work by Garon and Farinas,¹² another damage accumulation method based on summing the linearized damage is:

$$HI3 = C \left(\sum_{\text{inlet}}^{\text{outlet}} \Delta t \tau^{\beta/\alpha} \right)^\alpha \quad (5)$$

Recently, Grigioni *et al.*²¹ proposed a procedure in which they defined a mechanical dose D as:

$$D = t_{\text{exp}} \cdot \tau^{\beta/\alpha} \quad (6)$$

The time derivative of D was taken, and the damage accumulation was expressed as:

$$HI4 = \sum_{\text{inlet}}^{\text{outlet}} \alpha \cdot C \cdot \left[\sum_{j=1}^i \tau(t_j)^{\beta/\alpha} \cdot \Delta t_j + D(t_0) \right]^{\alpha-1} \cdot \tau(t_i)^{\beta/\alpha} \cdot \Delta t_i \quad (7)$$

The term in the square brackets represents the whole mechanical dose acting on the cell moving along the path line from the starting observation time until the i th instant, whereas $\tau(t_i)^{\beta/\alpha} \cdot \Delta t_i$ is the elementary mechanical dose sustained by the red cell in the i th interval. Like Equation 4, this equation reproduces Equation 1 for uniform shear stress; however, it also addresses the shear time-varying history sensed by the RBCs, which could not be achieved with Equations 3 and 4.

Similarly, another approach that takes into account the time-dependent shear loading history was proposed by Goubergrits and Affeld.²⁵ They assumed that the blood damage contribution from a time instant was a function of the blood damage at that time but was independent of how the existing blood damage arose. They define an effective time, t_{eff} as a virtual time step, and calculate HI accordingly as:

$$HI5(t + \Delta t) = C \cdot (t_{\text{eff}} + \Delta t)^\alpha \cdot \tau(t + \Delta t)^\beta \quad (8)$$

where t_{eff} is calculated as:

$$t_{\text{eff}} = \left(\frac{HI5(t)}{C \cdot \tau(t + \Delta t)^\beta} \right)^{1/\alpha} \quad (9)$$

Although hemolysis has been extensively studied both experimentally and computationally, there is still a debate on determining the most appropriate and universal modeling approach. Furthermore, there is less evidence of experimentally verified computational models. In this article, the HI predictions of five Lagrangian approaches ($HI1$, $HI2$, $HI3$, $HI4$, and $HI5$) and the Eulerian method with scalar transport (E), were evaluated and compared with measurements in two physical devices. The HI calculations were performed considering three different sets of constants for the power law equation as given in **Table 1**. The computational evaluation was carried out in an axial shearing device, which has a uniform shear region in the narrowest gap between the rotating spindle and the outer housing surfaces. This satisfies the experimental conditions of Equation 1. The evaluation was further extended to a clinical VAD in which there are complex flow and shear fields.

Materials and Methods

Blood Shearing Device

Blood Shearing Device (Hemolyzer-H). The blood shearing device¹⁰ for high-shear stress and short exposure time conditions (named “Hemolyzer-H”) was custom designed based on a modification of the Jarvik 2000 blood pump. It consists of a rotating rotor spindle, instead of the impeller with blades,

and housing without the diffuser. The computational model of the device consists of the spindle and the outer housing only (**Figure 1A**). The smallest gap between the rotor middle surface and outer housing is 0.1 mm. When the spindle is rotated, a uniform high-shear stress region is generated in this gap.

CentriMag Blood Pump. The CentriMag blood pump is an extracorporeal circulatory support device providing hemodynamic stabilization in patients in need of circulatory support. It has a magnetically levitated centrifugal pump impeller and has been used clinically worldwide.¹⁴ The computational model of the CentriMag blood pump consists of the inlet, the impeller, the outlet, and the outer casing wall (**Figure 2A**). There is a gap separating the rotor of the impeller and the housing wall.

Meshing

The 3D geometry of the Hemolyzer-H and the CentriMag blood pump were generated using the 3D computer-aided design software package SolidWorks v2007 (SolidWorks Corporation, Concord, MA). The geometries in Parasolid format were imported into the grid-generation software Gambit v.2.3.16 (Ansys Fluent Inc., Lebanon, NH). The flow domain of the Hemolyzer-H was meshed using 2.7 million hexahedral elements.

The axial position of the magnetically levitated impeller within the CentriMag pump is determined by the balance between magnetic and lift forces. This position was measured experimentally for each operating condition, and separate meshes were created for each condition. Each mesh used the same controlled hybrid (hexahedral and tetrahedral) structure (**Figure 2**). Hexahedral cells were used for the inlet and outlet, core, and gap regions. The rest of the geometry was meshed

with uniform size (3 mm) tetrahedral elements. The overall size of the model was 3 million cells.

Flow Calculations

The computational domain of the Hemolyzer-H is composed of the fluid, spindle surface, and the inner surface of the outer housing. No-slip boundary conditions were applied at the walls: rotational speed was specified for the spindle and zero velocity for the outer wall. Uniform velocity inlet and outflow outlet conditions were used and the inlet and outlet were extended so the solution was not constrained by these boundary conditions. Reynolds number (Re) was 20–220 at the inlet and 85 in the gap; therefore, laminar solutions were calculated.

The computational domain of the CentriMag blood pump is composed of fluid, the impeller/rotor surface, and the inner surface of the outer housing. The fluid domain was divided into two sections: rotating fluid and stationary outer fluid. The rotational motion of the impeller/rotor section was defined with the multiple reference frame (MRF), which is a steady-state approximation of the rotational motion. No-slip boundary conditions were applied at the walls, uniform velocity at the inlet, and uniform pressure at the outlet. Re at the inlet was 2100–5000, and therefore turbulent solutions were simulated. The SST- $k\omega$ turbulence model was used as it has been found to give good predictions of the pressure head in a similar centrifugal pump.¹⁵

The commercial finite-volume CFD solver Fluent (Fluent v6.3.26, Ansys Fluent Inc., Lebanon, NH) was used to solve the flow field. The semi-implicit method for pressure-linked equations pressure-velocity coupling algorithm was used, and all equations were discretized with second order accuracy. Flow convergence was monitored with the pressure head values as

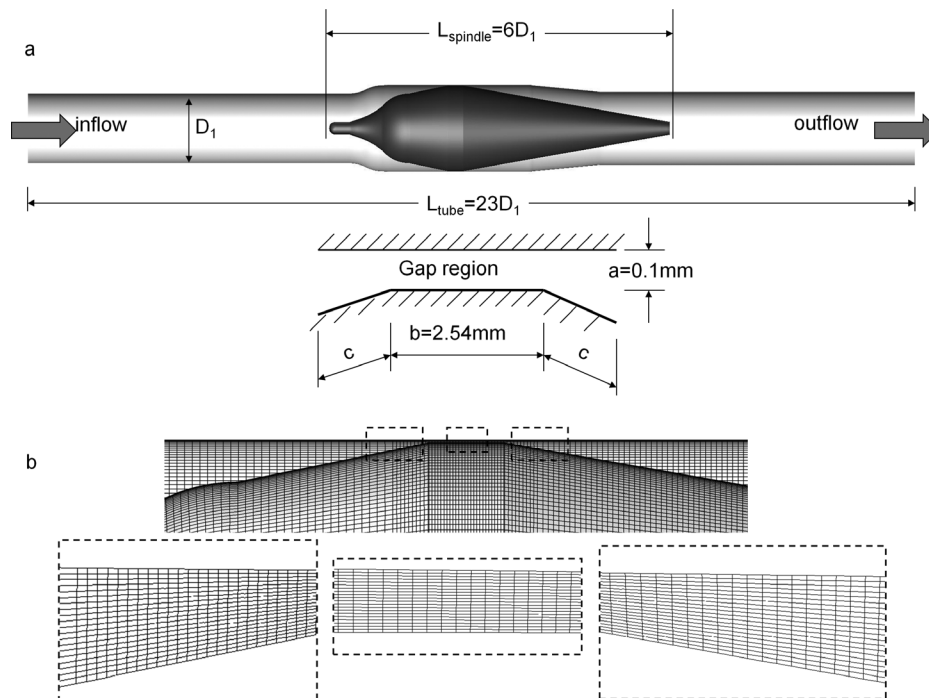


Figure 1. A: Flow domain and dimensions of Hemolyzer-H, and **(B)** sample mesh structure. D_1 , tube diameter; a , gap thickness; b , gap length; c , length of pre- or postgap region.

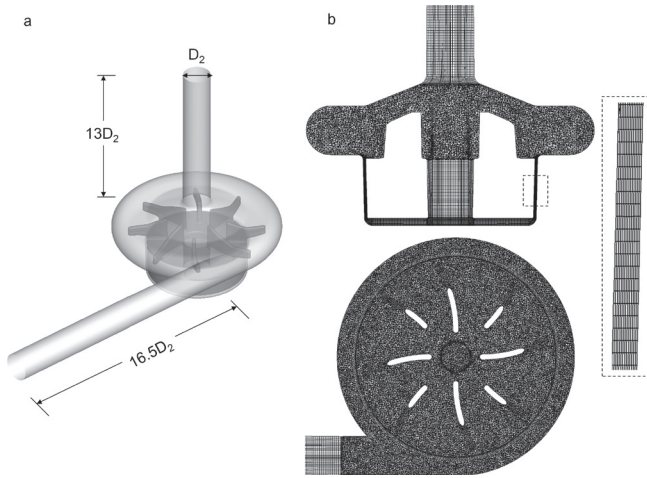


Figure 2. A: Model of the CentriMag blood pump, and **(B)** mesh structure. D_2 , tube diameter.

well as the continuity and momentum residuals and mass balance. For convergence, the pressure head values were stabilized, the velocity residuals were below 2×10^{-3} , and the mass balance errors were always $<3 \times 10^{-4}\%$ of the total mass flow rate. Blood was treated as an incompressible Newtonian fluid in our analyses. Although it is well known that blood exhibits non-Newtonian fluid behavior under low shear rate, it is reasonable to use the Newtonian assumption because shear rates in typical rotary blood pumps are relative high ($>100 \text{ s}^{-1}$).²⁶

The 3D shear stress tensor was derived from the velocity field, and a user-defined code was used to calculate the scalar shear stress²⁶ from the stress tensor components:

$$\tau = \left[\frac{1}{6} \sum (\tau_{ii} - \tau_{jj})^2 + \sum \tau_{ij}^2 \right]^{\frac{1}{2}} \quad (10)$$

Reynolds shear stresses were not included in the analyses because it has been shown that including them creates numerically elevated shear fields.^{27,28}

Hemolysis Calculations

The Eulerian hemolysis model was programmed and compiled into the CFD solver using user-defined functions and user-defined scalars (UDS). Once the flow solution was obtained, the scalar transport equation was solved. Convergence was monitored by the UDS residuals ($<10^{-7}$) and stabilized predictions at the exit of the device. The HI values for each computational cell were calculated by postprocessing the data with a separate user-defined code. To obtain the final HI, the mass-weighted average of the values at the exit of the device was calculated.

For the Lagrangian models, the path lines were obtained by postprocessing the flow solutions. Parameters for calculating the path lines were chosen so as to get as many path lines to reach the exit as possible. There were between 68 and 1974 path lines for the Hemolyzer-H and between 1165 and 1456 for the CentriMag blood pump. The HI calculations used Matlab (Matlab 6.5.1, The MathWorks Inc. Natick, MA). All the computations were carried out on a PC workstation with two Intel Xeon quad-core 2.66GHz CPUs and 16.0 GB RAM with Windows XP 64 bit operating system.

Experiments

Plasma hemoglobin was measured after a single passage through the Hemolyzer-H. The experiments were performed in the same way as those described in detail previously.¹⁰ Measurements of plasma hemoglobin in the CentriMag blood pump were carried out over a 3 hour period using a recirculating flow loop according to the recommended practice for assessment of hemolysis in continuous flow pumps specified by the American Society of Testing and Materials (ASTM F1841-97).²⁹ The following equation was used to find the single-pass HI from the change in plasma free hemoglobin over multiple passes:

$$HI(\%) = \frac{\Delta Hb}{QT} \frac{Hb}{V} \times 100 \quad (11)$$

where Q is the flow rate in the recirculating loop, T is the time the blood has been circulating, and V is the volume of blood in the loop.

Results

Hemolyzer-H Model

Mesh refinement. The mesh refinement study was performed with four different meshes of the Hemolyzer-H model, with a refinement ratio of two (**Table 2**). Convergence in the pressure drop with increasing mesh density was clear (**Figure 3A**). Meshes were evaluated according to the quantification of uncertainty principles outlined by Roache.^{30,31} The order of convergence was calculated:

$$p = \frac{\log \left(\frac{|\phi_{coarse} - \phi_{medium}|}{|\phi_{medium} - \phi_{fine}|} \right)}{\log(r_r)} \quad (12)$$

where ϕ_{coarse} , ϕ_{medium} , and ϕ_{fine} are the values of the parameter ϕ obtained using coarse, medium, and fine meshes, respectively, and r_r is the refinement ratio (two in our case). ϕ was the area-weighted average of the local shear stresses on the spindle. Meshes one, two, and three were used for the fine, medium, and coarse meshes, respectively. p was calculated as 0.95, showing that the convergence was around first order accurate. The following error estimates were used to calculate the mean percentage errors for each mesh (**Figure 3B**):

$$E_{fine} = \frac{|\phi_{medium} - \phi_{fine}|}{|\phi_{medium}|} \frac{1}{r_r^p - 1} \quad (13)$$

$$E_{medium} = \frac{|\phi_{coarse} - \phi_{medium}|}{|\phi_{coarse}|} \frac{1}{r_r^p - 1} \quad (14)$$

$$E_{coarse} = \frac{r_r^p |\phi_{coarse} - \phi_{medium}|}{|\phi_{coarse}|} \frac{1}{r_r^p - 1} \quad (15)$$

Mesh two was selected for the study, as the error was acceptable (5%), and the mesh size was manageable (<3 million cells).

Table 2. Mesh Details of the Hemolyzer-H Models

Mesh	Size (Million Cells)	Number of Nodes			Refinement Factor
		a	b	c	
1	21.66	40	200	80	2
2	2.71	20	100	40	2
3	0.35	10	50	20	2

a, gap thickness; b, gap length; c, length of pre- or postgap region.

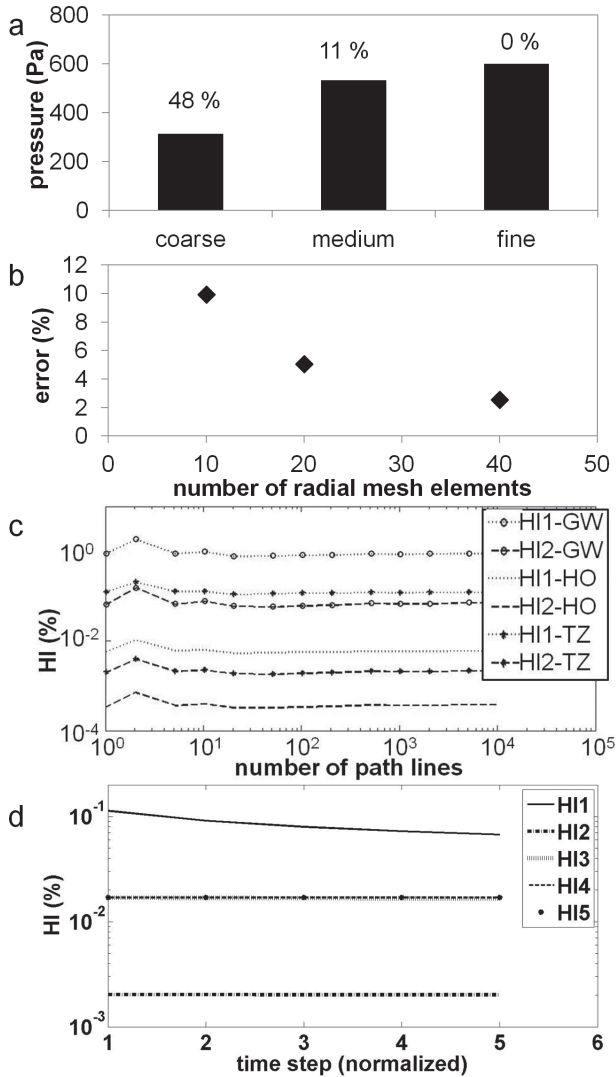


Figure 3. Hemolyzer-H model mesh refinement and path line dependence results. **A:** Pressure drop predictions. The pressure drops are the difference between pressure at the inlet and the outlet. The percentages shown on top of each bar are the deviations from the finest model (Model-1) predictions. Details of the meshes are given in Table 2. **B:** Standardized percentage error compared with the estimated continuum case. **C:** Variation in HI estimation with number of path lines. Note that for clarity, results are only shown for two accumulation methods for each set of power law constants. However, all five accumulation methods showed very similar variations. **D:** Variation in HI with size of time step. Results are normalized to the value used for the rest of the analysis, which had mean = 0.57 msec. HI, hemolysis index.

Shear stress field. The uniformity of the shear stress field in the gap region of the Hemolyzer-H was assessed by examining shear stress as contour plots and along path lines (**Figure 4**). The CFD-predicted mean scalar shear stress in the gap region varied between 203 and 208 Pa, as the flow rate was increased from 10 to 100 ml/min, which is in good agreement with the analytical value (198 Pa), calculated using the rotational speed, radius of the spindle at the gap region, fluid viscosity, and the thickness of the gap ($\tau = \omega \cdot r \cdot \mu / a$). At 400 ml/min the CFD-estimated mean scalar shear stress was 264 Pa. The difference between computational and analytical values is caused by the axial component of velocity, which creates a parabolic profile imposed on the larger linear profile and results in slightly higher shear stresses at the walls.

Small regions of high-shear stress were found upstream and downstream of the gap. The shear stress in these spots is significantly higher than the shear stress in the gap (up to 338 Pa). However, as their volumes are localized and extremely small, their effect on the total blood damage is assumed to be small. Shear stress along the path lines (**Figure 4B and C**) is higher inside the gap region than upstream or downstream; however, there are spikes in the shear stress plots in the upstream and downstream regions because of the presence of the high-shear spots. At 100 ml/min the path lines spent an average 1.3 msec within these high-shear spots compared with 17.6 msec in the gap.

HI Predictions. The HI predictions (**Figures 5 and 6**) from six different calculation approaches (five Lagrangian and the Eulerian) each with three sets of power law constants, along with the experimental results, are presented in **Figure 6** for varying flow rate conditions at a constant rotational speed of 8000 rpm. The calculations with GW constants overestimated for every HI calculation method, but were closest with HI2 (mean absolute error compared with experiments 0.24%, percentage error 1360%) compared with experimental values. Note that many predictions exceeded the scale of the graph. With HO constants, the predictions with the Eulerian, HI3, HI4, and HI5 methods all underestimated the experimental results whereas the HI1 approach slightly overestimated the results. With TZ constants HI2 predictions were closest but underestimated the experimental results (mean absolute error compared with experiment is 0.013%, percentage error 54%); whereas Eulerian, HI3, HI4, and HI5 methods overestimated the results.

CentriMag Blood Pump Model

The HI calculation approaches were further evaluated in the CentriMag centrifugal blood pump. **Figure 7** shows path lines through the CentriMag pump and variation in shear stress with exposure time along two sample path lines under the operating condition of 4000 rpm and 5 L/min. The non-uniform shear stress along the path lines is clear. **Figure 8** shows HI contours using the Eulerian model on vertical and horizontal cross-sectional slices through the CentriMag pump at a flow rate of 5 L/min and rotational speeds of 3000 and 5000 rpm. The increase in the rotational speed from 3000 to 5000 rpm caused a large increase in HI. A vector plot of an enlarged region at 5000 rpm shows the complexity of the flow field and how regions of high HI can develop. The HI predictions were compared with measurements at varied flow rates (3–7 L/min) with a rotational speed of 4000

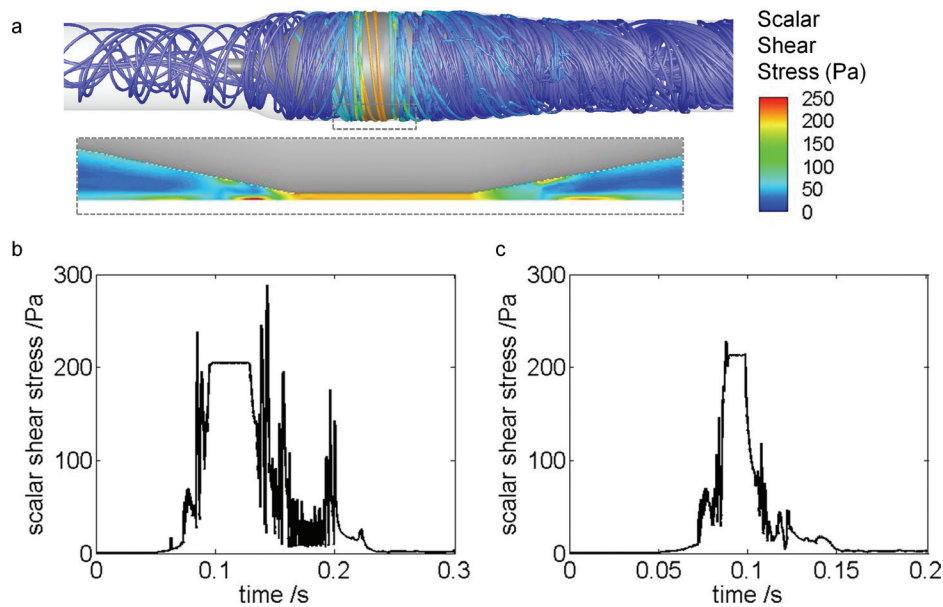


Figure 4. A: Illustrative path lines in Hemolyzer-H, and enlargement showing scalar shear stress contours in the gap region. Shear stress versus exposure time variations along sample path lines for the rotational speed of 8000 rpm and the flow rates of (B) 20 ml/min, and (C) 50 ml/min. The gap exposure times at these conditions are 33.8 msec and 13.5 msec, respectively. Note that the path line plots only show portions of the entire flow time.

rpm (Figure 9A), and for varied rotational speeds (3000–5000 rpm) at a flow rate of 5 L/min (Figure 9B). Using TZ constants the HI2 method gave predictions closest to experiments (mean absolute error 0.00034% and percentage error 116%). The mean absolute error with the Eulerian method was 0.0074% (percentage error 2279%). The closest with HO constants were HI3, HI4, and HI5 (mean absolute errors 0.00018%, 0.00020%, and 0.00020%, percentage errors 40%, 43%, and 42%) and Eulerian (mean absolute error 0.00026% and percentage error 101%).

Discussion

Reliable prediction of hemolysis is a key step to designing more biocompatible blood-contacting devices. In this work, we compared one Eulerian approach and five Lagrangian approaches for CFD-based prediction of hemolysis in

blood-contacting devices. We also compared the impact of three different sets of power law constants reported in the literature. The numerical predictions were compared against experimental measurements conducted on a custom-made blood shearing device with a uniform high-shear stress field, and a clinical VAD with a complex shear stress field. The mesh convergence studies and comparison of the pressure differences across the two devices showed that the calculations were satisfactory for the analysis presented given the limitations that are outlined in the ensuing sections. The mean error, mean percentage error, and correlation coefficient for each model and the combined data from both devices, are given in Table 3.

Eulerian Calculations

The Eulerian method for calculating hemolysis has an advantage over the Lagrangian methods because unlike the Lagrangian methods, which rely on seeded path lines to sample the flow, the whole fluid domain is included in the calculation. For the Hemolyzer-H device, the HI contours obtained at different flow rate conditions, with the TZ constants, are shown on the cross-sectional view planes in Figure 5.

The CentriMag HI contours, obtained for 3000 and 5000 rpm rotational speeds at 5 L/min flow rate, are shown in Figure 8. Again only the calculations with the TZ constants are presented. The highest concentration of plasma hemoglobin is around the rotor surface and at the bottom of the device, which is a result of the secondary flow path. Elevated hemoglobin concentration is also seen close to the inclined top outer housing wall, which is caused by a recirculation zone there (Figure 8, vector plot).

The differences between the HI values predicted by the Eulerian method and the experimental values for the CentriMag

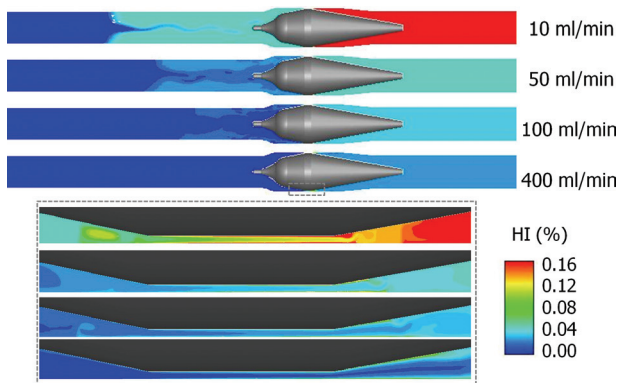


Figure 5. Hemolysis index (HI) contours using the Eulerian model on a cross-section through the Hemolyzer-H with an enlargement of the gap region. The rotational speed was 8000 rpm, and constants were TZ. TZ, Zhang *et al.* constants.

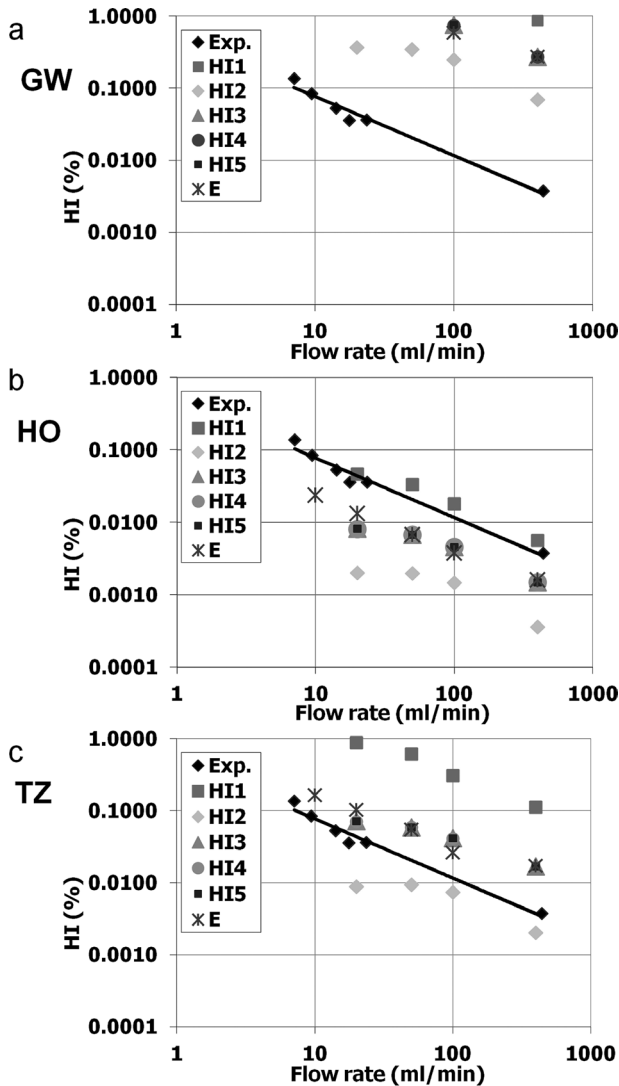


Figure 6. Hemolysis index (HI) measurements and predictions for the Hemolyzer-H at different flow rate conditions for different sets of power law constants: (A) GW constants, (B) HO constants, and (C) TZ constants. Note the logarithmic scale that highlights the order of magnitude of the differences and that *HI3*, *HI4*, and *HI5* are in almost identical positions. GW, Giersiepen and Wurzing constants; HO, Heuser and Opitz constants; TZ, Zhang *et al.* constants.

blood pump is higher than that for the Hemolyzer-H. This could be because of the complicated flow in the CentriMag blood pump with back flow and secondary flow paths. This complexity is a large deviation from the uniform shear stress conditions in the experiments used to obtain all sets of constants.

Lagrangian Calculations

The Lagrangian path lines were obtained by releasing pseudoparticles from the inlet surface of the corresponding device. The HI accumulations were calculated along the particle trajectories or path lines to the exit of the device. The number of path lines varied with the device and operating condition. The Hemolyzer-H with a flow rate of 20 ml/min used the fewest path lines, 68. To check whether this

was enough, we performed all the same HI calculations with half of these path lines and found percentage differences, as compared with the original HI predictions, of between 0.8 and 3.2%. Also, the variation in the calculated HI in the Hemolyzer-H with number of path lines was assessed up to 10^4 paths for one operating condition (Figure 3C), and the calculations were found to be independent of the number of path lines above 20 for this relatively simple device. However, a similar analysis on the CentriMag, with path lines up to the original number used, showed that at least 500 were required.

As stated by Grigioni^{21,32} integrating a mini power law does not reproduce the uniform shear stress results, and therefore *HI1* was not expected to produce accurate HI predictions. In fact, the *HI1* results with HO constants have small errors and fairly large correlation coefficients (>0.95). However, results for the *HI1* depended on the size of the time steps that the path line was divided into (Figure 3D), whereas the other Lagrangian methods were independent of the time step. Had a different time step been used for the analysis with the *HI1*, the results would be different.

The calculated HI values with the *HI3* accumulation method using HO constants provided the smallest relative error (57%). However, on average, the *HI2* accumulation method had the smallest relative error compared with the experimental values (as compared with the averages of the other methods), but had the smallest correlation coefficients between the predicted results and experimental values with all three sets of constants. It can be shown that although the *HI2* accumulation method is correct for a constant uniform shear stress it is unlikely to be correct for anything else. For example, in a case where a constant shear stress is applied for a fixed time interval, and at all other times there is no stress, *HI2* will depend on the actual time at which the shear stress interval starts. Because of the time exponent ($\alpha-1$) in the equation, which is negative for fractional α , the later the shear stress is applied the lower the predicted hemolysis. This explains why, even for the hemolyzer, in which the stress is close to uniform, the *HI2* results are much lower than those of *HI3*: the stress occurs in the middle of the device rather than at the start.

The HI values from *HI3* were not influenced by the time step size, or the actual time, and therefore *HI3* looks a promising Lagrangian method. However, the correlation coefficients are below 0.9 and significantly lower than the Eulerian method. Both *HI3* and the Eulerian method are based on the same damage equation, only the implementation differs. We believe the path lines do not fully represent the complicated flow domains studied here. For example, path lines initiated from the inlet will never enter a recirculation zone, whereas those seeded in a recirculation zone will never exit the device.

The HI values calculated with *HI4* and *HI5* prediction methods, although not exactly the same, were very close to each other, and to *HI3*. Both *HI4* and *HI5* methods aim to account for the loading history. The reasoning is that exposure to high-shear stress at a previous time step may result in higher damage at the current time step, as compared with exposure to low shear stress in the previous time step and the same stress in the current time step. Although the two methods differ in their

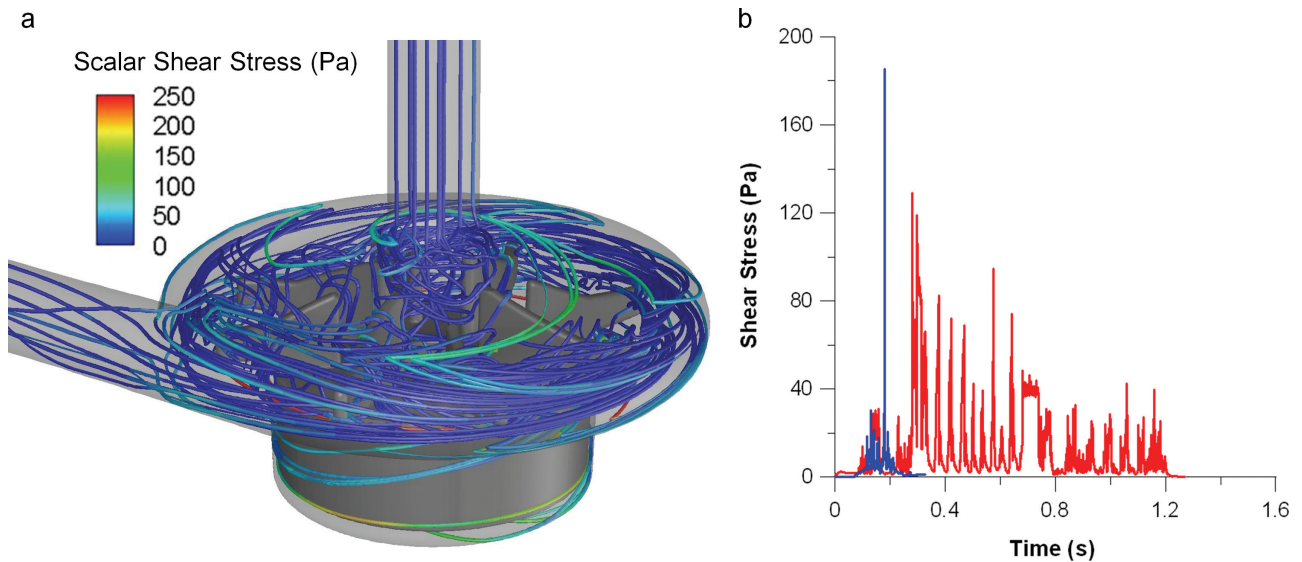


Figure 7. A: Illustrative path lines through the CentriMag pump and **(B)** variation in shear stress with exposure time along two sample path lines. The operating condition was 4000 rpm and 5 L/min.

implementation schemes, the results for the cases studied here are almost identical. They are also very close to the results for *HI3*, showing that there was no advantage to incorporating the damage history. In fact, the correlation coefficients for *HI4* and *HI5* were lower than those for *HI3*.

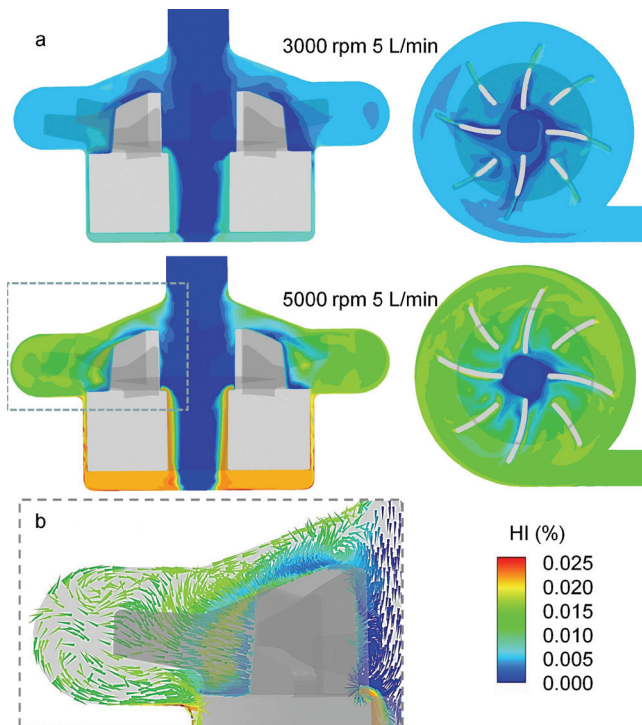


Figure 8. A: Hemolysis index (HI) contours using the Eulerian model on vertical and horizontal cross-sectional slices through the CentriMag pump at a flow rate of 5 L/min and rotational speeds of 3000 and 5000 rpm. **B:** A vector plot of an enlarged region at 5000 rpm shows the complexity of the flow field and how regions of high HI can develop. For clarity, not all vectors are shown. The TZ constants were used for the calculations shown. TZ, Zhang *et al.* constants.

Limitations

The experimental measurements of hemolysis presented here used ovine blood. However, the fragility of RBCs from different species is likely to differ^{33,34} so this validation analysis is valid only for ovine blood. In addition, the analysis is limited to Couette-type flows and the flow through a specific clinical VAD. It is expected that the errors would be similar in other similar devices, that is, in other centrifugal VADs, however, further studies are required to extend the analysis for other blood-contacting devices in which the shear stresses and exposure times are significantly different.

Blood was treated as a single phase Newtonian fluid. Shear rates were high ($>100 \text{ sec}^{-1}$) in the vast majority of the blood volume (the CentriMag pump has $<2\%$ of the volume with a shear rate $<100 \text{ sec}^{-1}$), which justifies neglecting the shear thinning nature of blood. However, blood consists of two major phases: plasma and RBCs. In certain situations, the relative concentrations of the two phases varies, for example, in narrow capillary tubes. The gap in the hemolyzer was just 100 μm , so the concentrations may vary across the gap. The models presented cannot account for this.

The MRF method was used to account for the rotation of the CentriMag pump impeller. The MRF method is an approximation that cannot account for transient effects or conditions other than steady state. In reality, as the impeller blades pass the outlet at the cut-water region there is a fluctuation in the pressure head and the flow field, and hence the shear stress field changes.

This work assesses the hemolysis models for reproducing *in vitro* hemolysis measurements and does not attempt to account for the complicated nature of flow *in vivo*. When VADs are implanted, the flow through them is determined by the interaction of the VAD and the native cardiovascular system and therefore has a pulsatile component. *In vivo*, the plasma free hemoglobin is a balance between the amount of hemoglobin coming from the RBCs and the amount being processed by the kidneys. These hemolysis models cannot calculate the *in vivo* hemolysis.

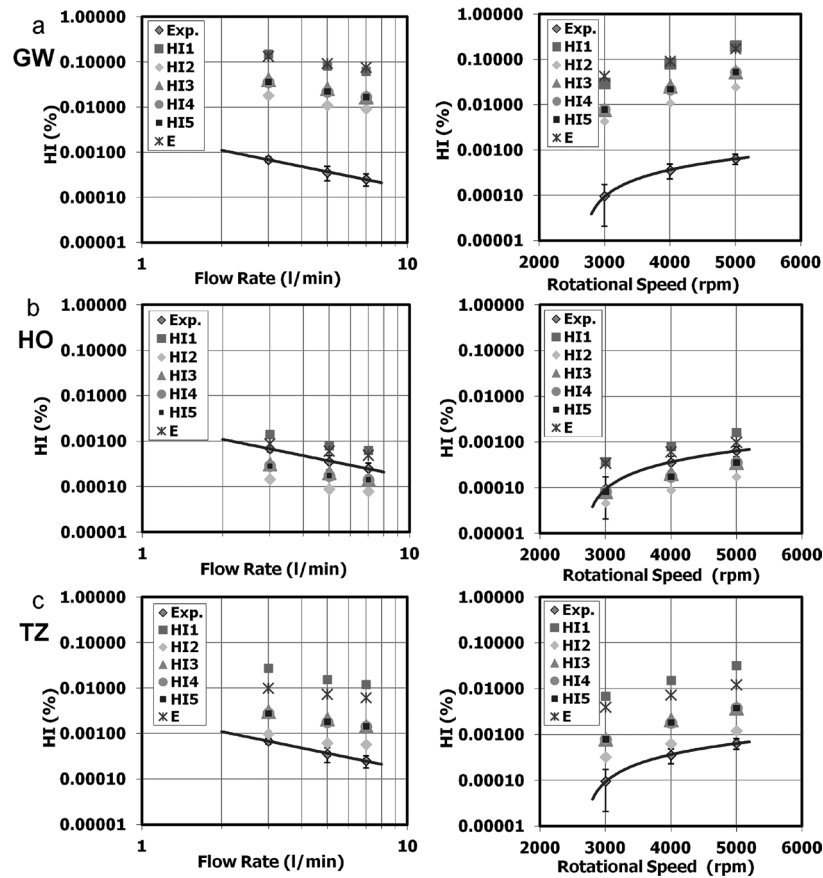


Figure 9. Variation in hemolysis index (HI) measurements and predictions in the CentriMag pump with flow rate and rotational speed for the range of models and with the three sets of power law coefficients (A) GW, (B) HO, and (C) TZ. For varying flow rate, the rotational speed was fixed at 4000 rpm and for varying speed the flow rate was fixed at 5 L/min. The experimental values are the averaged measurements, and experimental deviations are indicated with the upper and lower error bars. Note the logarithmic scale that highlights the order of magnitude of the differences and that *HI*3, *HI*4, and *HI*5 are in almost identical positions. GW, Giersiepen and Wurzing constants; HO, Heuser and Opitz constants; TZ, Zhang *et al.* constants.

Table 3. Comparison of Predicted HI Values, from Each Hemolysis Calculation Method and With Each Set of Constants, and the Experimentally Measured HI Values

Constants	Method	Error	% Error	Correlation Coefficient
GW	HI1	1.672	21366	0.957
	HI2	0.114	2327	0.819
	HI3	0.403	5805	0.893
	HI4	0.400	5664	0.893
	HI5	0.402	5695	0.893
	E1	0.775	15335	0.996
HO	HI1	0.003	110	0.952
	HI2	0.008	81	0.793
	HI3	0.007	57	0.898
	HI4	0.007	58	0.883
	HI5	0.007	58	0.883
	E1	0.011	91	0.999
TZ	HI1	0.215	3479	0.964
	HI2	0.006	94	0.741
	HI3	0.013	350	0.877
	HI4	0.013	337	0.875
	HI5	0.013	342	0.875
	E1	0.025	1106	0.995

Data are from all the operating conditions for both Hemolyzer-H and CentriMag blood pump.

HI, hemolysis index, GW, Giersiepen and Wurzing constants; HO, Heuser and Opitz constants; TZ, Zhang *et al.* constants.

Conclusions

The comparative numerical and experimental results presented here demonstrate clearly that the accuracy of power law-based hemolysis prediction is influenced by both the power law constants and the numerical approach adopted. None of the methods could reproduce the experimental results satisfactorily. The Eulerian scalar transport methods, although having considerable errors compared with the experiments (minimum percentage error = 91%), had high correlation coefficients (>0.99), suggesting that this approach can be used to give relative comparisons of hemolysis in different devices, which would be useful, for example, for ranking different devices or for design optimization.

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